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Academic Session: 2023-24

Assignment

B.Tech. IV Sem Section: M3 MTH-241

1. A cement company has three factories which manufacture cement which is then transported to four distribution centres. The quantity of monthly production of each factory, the demand of each distribution centre and the associated transportation cost per quintal are given below:

Factory	Distribution Centres				Monthly production (in quintals)
	W	X	Y	Z	
A	10	8	5	4	7,000
B	7	9	15	8	8,000
C	6	10	14	8	10,000
Monthly demands (in quintals)	6000	6000	8000	5000	25,000

- (a) Suggest the optimum transportation schedule.
 (b) Is there any other transportation schedule which is equally attractive? If so, write that.
 (c) If the company wants that at least 5,000 quintals of cement are transported from factory C to distribution centre Y, will the transportation schedule be any different? If so, what will be the new optimum schedule and the effect on cost?

2. A truck company on a particular day has 5 trucks for sending material to 6 terminals. The cost of sending material from some destination to different trucks will be different as given by the cost matrix below. Find the assignment of 4 trucks to 4 terminals out of the six at minimum cost.

Terminals	Trucks				
	A	B	C	D	E
1	3	6	2	6	5
2	7	1	4	4	7
3	3	8	5	8	3
4	6	4	3	7	4
5	5	2	4	3	2
6	5	7	6	2	5

3. Five operators have to assigned to five machines. The assignment cost are given in the table below:

Operator	Machine				
	I	II	III	IV	V
A	5	5	2	2	6
B	7	4	2	3	4
C	9	3	5	-	3
D	7	2	6	7	2
E	6	5	7	9	1

Operator A cannot operate machine III and operator C cannot operate machine IV. Find the optimal assignment schedule.

4. Solve the game whose payoff matrix is given below:

		Player B		
		B_1	B_2	B_3
Player A	A_1	-3	-1	6
	A_2	2	0	2
	A_3	5	-2	-4

5. Solve by the following game by using L.P process,

$$A \begin{bmatrix} 3 & 2 & 4 & 0 \\ 3 & 4 & 2 & 4 \\ 4 & 2 & 4 & 0 \\ 0 & 4 & 0 & 0 \end{bmatrix}$$

8 x 5